

since there are different sub-divisions in the field. For example, mission computer and related hardware, cockpit displays, inertial navigation systems, weapons guidance system, etc., are parts of hardware. (Providing) software for those systems is again a challenging task,” says Suresh Kumar, Director-Sales, Datasol (B), which designs, develops and integrates modular products focusing on the areas of automated test equipment, avionics and embedded Systems.

The different sub-areas you can specialise in avionics are flight control system, flight management system, communication system, navigation system, surveillance system, safety and security system, power and electrical control system, in-flight entertainment and other miscellaneous systems.

On The Radar

You can decide to work either on the software side or the hardware side of avionics systems. Besides defence, options of working with other services-oriented organisations like the IT companies as well as the maintenance, repair and overhauling companies are widely open.

IT companies provide solutions on both hardware as well as software side for working with avionics systems.

On Hardware side

Detailed hardware design (VLSI-digital and analog, FPGA, ASIC), testing of the equipments and components like radio frequency chips, quality and qualification testing with environment, system safety analysis and even critical hardware manufacturing and development are a few options. IT solutions organisations are involved in providing the DTI (Design, Testing and Integration) solutions to airlines.

Today, the latest aircraft carry a unified solution in form of an open architecture computer as well. By synchronising the other subsystems like the radar systems and sensory systems with the core avionics systems, mission critical data is processed and displayed. The same data is communicated through radio communication to specified towers at airports.

Electronics engineers get involved mostly with designing and testing of the avionics hardware and subsystems. For example, VLSI chip design engineers specialise in designing different silicon components for mission computers based on open architecture.

On software side

Mission-critical electronic systems interface with the hardware using some software systems. The computers and embedded electronics used in aircraft today run on real-time operating systems. It’s the data processing software that collects the data from the mission computer based on open system standards.

Avionics software and components testing professionals are expected to check whether different avionics subsystems and hardware components are working properly or not. In-flight entertainment services also require



“Understand the workings and develop a passion for Aircraft instrumentation and Engineering. Avionics jobs are similar to other engineering jobs, with the exception of need for aircraft knowledge.”

C.V. Ravi Shankar
Technical Director
Merlinhawk Aerospace
Pvt. Ltd.

strong networking and telecommunication support, which has to match the regulatory standards. Every organisation working with hardware and software of avionics systems must meet the respective standards for hardware and software published by the Radio Technical Commission for Aeronautics (RTCA).

For software programmers, knowledge of different basic programming languages like C, C++, MATLAB, and familiarity of working with RTOS as well as other platforms is necessary. Simulated environments are created to test avionics systems. Proficiency or awareness of CAD / CAE / CAM design tools and flair for design is also expected—it’s been the most essential practice in the automobiles as well as aviation industry.

Take-off Conditions

Deep interest in using cutting-edge computing ability and advanced electronics in challenging applications areas form basic elements for avionics education.

A strong fundamental background in mathematics, physics, electronics, chemistry, computer science, mechanical drawing / drafting, and conventional software programming is also helpful.

“Usually candidates carrying bachelors / masters level engineering degree or aeronautical engineering or aircraft maintenance and engineering can have an edge over others by specialising in systems engineering and instrumentation engineering,” says Retd. Air Cmde. Raghubir Singh, Director-Training at the Indian Institute of Aeronautical Engineering and Information Technology, Pune.

“Certain universities like Anna University, VTU and IITs offer a specialisation in avionics. A career in avionics requires an ability to appreciate and understand systems engineering. In today’s world where avionics has graduated from a standalone cluster of systems like FCS, FMS, and CNS to a level where the latest concept is “Integrated Modular Avionics”—it is important that an aspirant has solid grounding in electronics and further systems engineering,” says Abhishek Vanamali, Group Manager-Marketing, HCL’s Aerospace Practice, HCL Technologies Pvt. Ltd., which offers specialised expertise for hardware and software system level solution services.

Many colleges / institutes tie up with foreign universities to offer specific and specialised courses. For instance, Sreenidhi Institute of Science and Technology, Hyderabad (accredited by AICTE) has tied up with Vaughn College of Aeronautics and Technology, New York, to provide courses for electronics engineering and technology with specialisation in avionics.

Find Your Way

The procedure of recruiting remains uniform for this industry—a written aptitude test followed by personal interviews. At times, private sector companies have practical exams to test the hands-on knowledge and strength of the candidates in their respective field of specialisation.

“If you’re really keen about avionics then you can take up research as your career goal and join research establishments like the Defence Avionics Research Establishment. You can also work for organisations like NAL, HAL, ISRO and DRDO or labs like ADE and ADA”, says Vanamali.

Gaining hands-on experience and undergoing training at the apprentice level is a crucial part of an avionics career.

Besides academic qualifications, other basic skills like job knowledge, reasoning power, analytical ability, task orientation, tool identification, innovation, communication and being a team player are expected in freshers.

“The best thing would be to have some industry participation with colleges during the project phase. More than seminars, one-on-one hands-on training programs may be more useful in minimising the demand-supply gap”, says Kumar.

Know Your Role

The levels of hierarchy in this sector remain the same as other occupations. The typical jump from the fresher level to the intermediate level requires good knowledge and rigorous hands-on working experience in dealing with mission-critical hardware and software.

One can expect to work in the following hierarchy:

- Top level-Director / Project Manager
- Middle Level-Avionics / Aircraft Engineer
- Intermediate-Senior Technician / Developer / Software Tester
- Entry Level-Asst. Engineers / Technicians
- With the IT outsourcing companies the structure is:
- Top level-Vice President / Director / Project Manager
- Middle-Technical Lead
- Intermediate-Senior Tools Developer / Designer / Architect / Tester
- Entry level-Programmer / Tool Developer / Designer

Note that here the structure remains more or less the same whether you work on hardware design-testing or software development.

At IT outsourcing companies, one month of vigorous training on avionics domain—software development tools and verification tools—is provided before candidates are deployed on projects. Due to shortage of skilled workforce, many IT outsourcing units have developed and implemented their own training programmes for fresh recruits. Generally, training involves developing familiarity of working with co-ordination between software and hardware systems.

The Future

Like every other industry in the IT and ITes domain, the avionics industry is also starved for a skilled work-

Indicative College List	
■ IIT-Kharagpur	■ Nehru College of Aeronautics & Applied Science, Coimbatore
■ IIT-Bombay	■ Park College of Engineering and Technology, Coimbatore
■ IIT-Kanpur	■ School of Aviation Science and Technology, New Delhi
■ IIT-Madras	■ Indian Institute of Aeronautical Science, Jamshedpur
■ Indian Institute of Aeronautical Engineering, Dehradun	■ Institute of Aviation & Technology, Bhubaneswar
■ Academy of Aerospace & Aviation, Indore	■ Nehru College of Aeronautics & Applied Science, Coimbatore
■ Bangalore Institute of Aeronautical Engineering & IT, Bangalore	■ P.A.M.E College, Patiala, (Punjab)
■ Hindustan College of Engineering Kelambakkam (Tamil Nadu)	■ Rajiv Gandhi Memorial College Of Aeronautics, Jaipur
■ Hindustan Electronics Academy, Bangalore	■ School of Aviation Science and Technology, New Delhi
■ Hindustan Institute of Aeronautics, Bhopal	■ VSM Aerospace, Chelekere, Bangalore
■ Hindustan Institute of Engineering Technology, Chennai	■ Buddha Institute of Aviation Technology, New Delhi
■ Indian Institute of Aeronautical Engineering and Information Technology, Pune	■ Indian Institute of Aeronautical Engineering, Dehradun
■ Indian Institute of Aeronautics, Patna	■ Indraprasth Institute of Aeronautics, Gurgaon
■ Institute of Aviation Technology, Haryana	■ Academy of Aerospace & Aviation, Indore
■ Madras Institute of Technology, Anna University, Chennai	■ National Defence Academy, Pune
■ National Institute of Aeronautical Engineering, Dehradun	

force. The private aviation sector is yet to make it big by constructing an entire plane (for civil or military use) or complete engine by itself. Machines are getting more automated as technology advances, so the scope for growth is high.

“With a number of regional airlines coming into existence and growing at rapid speed while other national airlines go international, the growth of employment opportunities will take place. The industry has bright prospects,” says Mehta.

The outsourcing business of the aviation industry has also gone up. “We’re already ahead in terms of software development and the research front. With the 30 to 50 per cent offset clause for aircraft and defence purchase, lot of joint ventures in aviation industries are coming up like maintenance and repair organisations at the Tier-I and Tier-II cities. Apart from that many industries and sub-industries will emerge with huge financial outlays,” says Singh.

Major developments are afoot in this sector. For example, DaimlerChrysler is setting up research centres, IT honchos like Infosys, TCS and HCL are offering specialised engineering and design services, and firms like Hindustan Aeronautics are planning to invest in R&D as well as manufacture aircraft components in India.

Patel has predicted at the World Economic Forum that the fleet of aircraft operating domestically is expected to increase from the current 400 to about 2,500 by 2020. Hence the long-term equation for the avionics sector is simple—more passengers, more planes, more demand for skilled work force. ■

samir.makwana@thinkdigit.com